

TASK 02 REPORT

INTERNSHIP: CYBERSECURITY

Name: Aneeqa Kamran

Course: Cyber Security

Student ID: DHC-8189

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1. Application Setup

A mock web-based application was set up locally using Node.js and React. The application was executed on `http://localhost:4100`. Key features such as user registration, login functionality, and profile access were explored to prepare for security testing.

2. Tools Used

- 1 OWASP ZAP for automated vulnerability scanning
- 2 Browser Developer Tools for manual inspection
- 3 Localhost environment for testing the application

3. Vulnerability Assessment

The OWASP ZAP tool was used to perform an automated scan on the running application. The traditional spider was selected to crawl the application without using browser automation. The scan identified multiple vulnerabilities related to missing security headers and configuration weaknesses.

4. Vulnerabilities Identified

- 1 Content Security Policy (CSP) Header Not Set – increases the risk of XSS attacks.
- 2 CSP Directive Missing – weakens overall security policy.
- 3 Cross-Domain Misconfiguration – improper CORS configuration allows unsafe access.
- 4 HTTP Only Site – communication is not encrypted.
- 5 Missing Anti-clickjacking Header – application vulnerable to clickjacking.
- 6 Subresource Integrity Missing – external resources are not verified.
- 7 Server Information Disclosure – exposes backend technology details.
- 8 Timestamp Disclosure – reveals system timing information.

5. Areas of Improvement

- 1 Implement HTTPS to secure data transmission.
- 2 Add Content Security Policy (CSP) headers.
- 3 Configure proper CORS settings to restrict access.
- 4 Hide server information in HTTP headers.
- 5 Add security headers such as X-Frame-Options and X-Content-Type-Options.
- 6 Ensure integrity checks for external resources.

Conclusion

The vulnerability assessment identified several medium and low-level security issues. While no critical vulnerabilities were detected, the identified weaknesses highlight the importance of implementing proper security configurations to protect web applications.